

Assessment 1 – Problem Statements

Course Code: CSA1704

Question 1: Water Jug Problem

Problem Statement

A person is provided with two water jugs of different capacities, one capable of holding 4 gallons of water and the other capable of holding 3 gallons. Neither jug has any measurement markings or indicators to show the amount of water contained within it. A water pump is available to fill either jug completely, and water can be emptied onto the ground whenever necessary. Water may also be transferred from one jug to another until either the source jug becomes empty or the destination jug becomes full. The challenge is to determine a sequence of operations that will result in exactly 2 gallons of water being present in the 4-gallon jug. The problem requires the application of logical reasoning and state-space search techniques to reach the desired goal state.

Question 2: Mars Rover Agent

Problem Statement

A Mars Rover is an intelligent robotic agent designed to explore the surface of Mars and perform scientific investigations without continuous human control. The rover gathers information about its surroundings using various sensors, cameras, and scientific instruments. It navigates through unfamiliar terrain, avoids obstacles, collects soil and rock samples, analyzes environmental conditions, and transmits valuable data back to Earth. Since the Martian environment is uncertain and dynamic, the rover must make decisions autonomously based on the information it perceives. Analyze the Mars Rover as an intelligent agent by identifying its percepts, describing its operating environment, listing the actions it can perform, determining suitable performance measures, and selecting the most appropriate agent architecture for efficient exploration and scientific discovery.

Question 3: 8-Queens Problem

Problem Statement

The 8-Queens problem is a classical artificial intelligence and search problem that involves placing eight queens on a standard 8×8 chessboard. The objective is to arrange the queens in such a way that no queen can attack any other queen according to the rules of chess. Since a queen can move horizontally, vertically, and diagonally, no two queens should occupy the same row, column, or diagonal. The problem requires exploring a large search space of possible board configurations and finding a valid arrangement that satisfies all constraints. Formulate the problem by defining the initial state, state space, operators, goal state, and cost function, and identify an appropriate search strategy that can efficiently find a solution.

Question 4: OLA Cab Booking Application

Problem Statement

An OLA cab booking application enables customers to book transportation services through a mobile or web-based platform. A customer enters the pickup and destination locations and selects a preferred cab type such as Mini, Micro, Sedan, Shared, or Prime. The system then checks the availability of drivers, estimates the fare, confirms the booking, and assigns a suitable driver to the customer. The application continuously tracks the journey until the passenger reaches the destination. The system must make decisions based on user requirements and available resources to achieve the customer's transportation goal. Analyze the OLA cab booking system as an intelligent agent and determine the type of agent involved, explaining how it helps achieve the desired objective.

Question 5: Uniform Cost Search (UCS)

Problem Statement

A logistics company manages a network of warehouses connected by transportation routes. Each route between warehouses has an associated travel cost that may represent fuel consumption, transportation charges, or delivery

time. The company aims to transport packages from a starting warehouse, denoted as S, to a destination warehouse, denoted as G, while minimizing the total transportation cost. Several alternative paths may exist between the source and destination, each with different cumulative costs. The challenge is to determine the most economical route using the Uniform Cost Search (UCS) algorithm. The problem involves exploring the weighted graph systematically and selecting the path with the lowest total cost while ensuring an optimal solution.

End of Assessment 1 Problem Statements